



POLITECNICO DI TORINO

MASTER'S DEGREE IN CYBERSECURITY ENGINEERING

Design and Implementation of an eBPF-based Security Monitoring Tool

Supervisor:

Prof. Riccardo Sisto

Candidate:

Simone Romano

Company Tutors:

Dr. David Soldani

Dr. Vincenzo Cosi

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This thesis presents the design and implementation of a security monitoring tool that collects and analyzes network connection data directly from the operating system kernel using eBPF. By leveraging kernel-level instrumentation, the system captures detailed information about network activity with high accuracy and minimal overhead, overcoming the limitations of traditional user-space monitoring approaches.

The proposed tool adopts a modular and efficient architecture to ensure scalability and real-time data collection. The gathered data is structured to support security analysis, enabling the detection of anomalous behaviors such as suspicious outbound connections, unauthorized communication patterns, and potential data exfiltration attempts.

This work demonstrates how kernel-derived network telemetry, combined with eBPF technology, can enhance system visibility and provide a solid foundation for advanced network security monitoring and threat detection.

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Acronyms

eBPF	Extended Berkeley Packet Filter
OS	Operating System
NIDS	Network Intrusion Detection System
CVE	Common Vulnerabilities and Exposures
OOB	Out Of Bounds
JIT	Just-in-Time

Chapter 1

Introduction

1.1 Context and Motivation

The extended Berkeley Packet Filter (eBPF) is a new programmability mechanism that changes the way we can observe and interact with the Linux kernel. Originally it was designed as a packet filtering mechanism, but now the paradigm has evolved to a more general purpose, addressing the need for observability and security in the Linux kernel.

eBPF introduces a tradeoff in kernel development: on one hand, it allows developers to extend the kernel with new features without the need to modify the kernel source code; but on the other hand, this widely adopted Linux kernel technology presents a new challenge for security, because it allows developers to run custom code in kernel space, which is a privileged environment.

When we use this mechanism to load custom programs in the kernel, we need to be sure that the code is safe and does not compromise the security of the system. eBPF addresses this problem by introducing a sophisticated verifier framework. This framework is able to analyze the code before it is loaded in the kernel, and to reject any code that does not respect the security and performance constraints. It permits programs running in the user space to submit eBPF bytecode to the kernel, which is then verified.

Once the programs are verified, they are executed in the kernel space, achieving high performance and low overhead through Just-in-Time (JIT) compilation, while maintaining or even enhancing integrity and security of the

kernel.

eBPF allows attaching eBPF programs to specific kernel hooks, enabling the observation and collection of events in the kernel space, without the need to modify the kernel source code.

This capability enables a new approach to security monitoring, allowing us to observe security-relevant events in a reliable and structured way, collecting them from different kernel hooks, transporting them to the user space, decoding them and making them usable for further analysis. As a result, the impact of this new technology is significant, because it allows us to build a complete pipeline from the kernel space to the user space.

Organizations leverage this new technology, because it permits to leverage the combination of safety, performance and observability, to build next-generation security tools that run in a privileged environment.

1.2 Thesis Objectives

This thesis aims to explore the potential of eBPF for security monitoring, by designing and implementing a system that leverages eBPF to collect relevant events from the kernel space, transport them to the user space, and finally make them usable for further security analysis.

In order to achieve this goal, the thesis will focus on the following specific objectives:

- Implement a kernel-side component that uses eBPF to attach to specific kernel hooks and collect raw events from the kernel space.
- Implement a user-space component that receives the raw events from the kernel space, decodes and enriches them, and finally outputs them in a structured format that can be used for further analysis.

1.3 Proposed Approach

1.4 Contributions

1.5 Thesis Structure

The thesis is structured as follows:

- **Introduction** introduces the context and motivation of the thesis, presents the objectives, the proposed approach, and the contributions of this work.
- **Background** provides a background on eBPF, its architecture, and its security implications.
- **System Design** describes the design and implementation of the proposed system for security monitoring using eBPF.
- **Evaluation** presents the evaluation of the implemented system, including performance measurements and security analysis.
- **Conclusion** concludes the thesis with a summary of the findings, limitations, and future work.

Appendix A

Appendix A

Appendix text here.

Appendix B

Appendix B

Appendix text here.

To my family and friends, for their support.